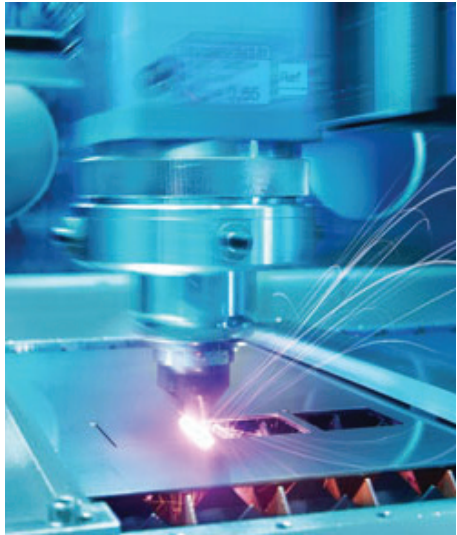


shells that have one or more nanometre-sized dimensions. For example, carbon nanotubes measure a diameter in the nanoscale, but can be several hundred nanometres long or even longer. Nanofilms or nanoplates have a thickness in the nanoscale, but their other two dimensions can be much larger.

The key is to be able to both, see and manipulate nanomaterials in order to take advantage of their special properties.



Working at the nanoscale

Manufacturing

Manufacturing at the nanoscale is known as nanomanufacturing. Nanomanufacturing involves scaled-up, reliable and cost-effective manufacturing of nanoscale materials, structures, devices and systems.

There are two basic approaches to nanomanufacturing: top-down and bottom-up.

- ▶ **Using the top-down method**, researchers selectively modify the starting material much like an artist creates a sculpture from a slab of marble. This method is more traditional, in that, material is altered (e.g. removed, added to, etc.) by mechanical or chemical means. Photolithography and electron beam lithography are examples of top-down approaches that are used extensively in the semiconductor industry to fabricate integrated electronic circuitry.
- ▶ **Using the bottom-up method**, researchers seek to build larger and more complex systems molecule by molecule. With this method, molecules are designed and created with the ability to spontaneously self-assemble when a chemical or physical trigger is applied. Nature routinely uses this method. Although bottom-up processes are less developed and understood, they hold great promise for the future.

Structures and properties of materials can be improved through these nanomanufacturing processes and can be made stronger, lighter, more durable,